SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether an external verifier accepted the result.

PUBLIC-GOODS REQUEST

$150,000

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

808
public GitHub stars

168
signed prospective cells

0
adversarial false passes

OPEN
MIT code and public proof

Prepared 2026-08-01. Dated GitHub interest signal; not user or installation counts. Not submitted.

PROBLEM

The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

01 **Declared plan**
Models, tools, topology, and limits

02 **Observed execution**
Calls, attempts, artifacts, and cost lenses

03 **External verdict**
Deterministic verifier outside the route

04 **Signed evidence**
Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN

Thesis: optimize the model-externally verified operation, not an isolated model call.

WORKING SUBSTRATE

Built before the grant.

Citadel starts with one command, then exposes stronger control only when the operation needs it.

/do — **Beginner path**

Natural-language routing, verification, and a durable next action.

PLAN — **Portable contract**

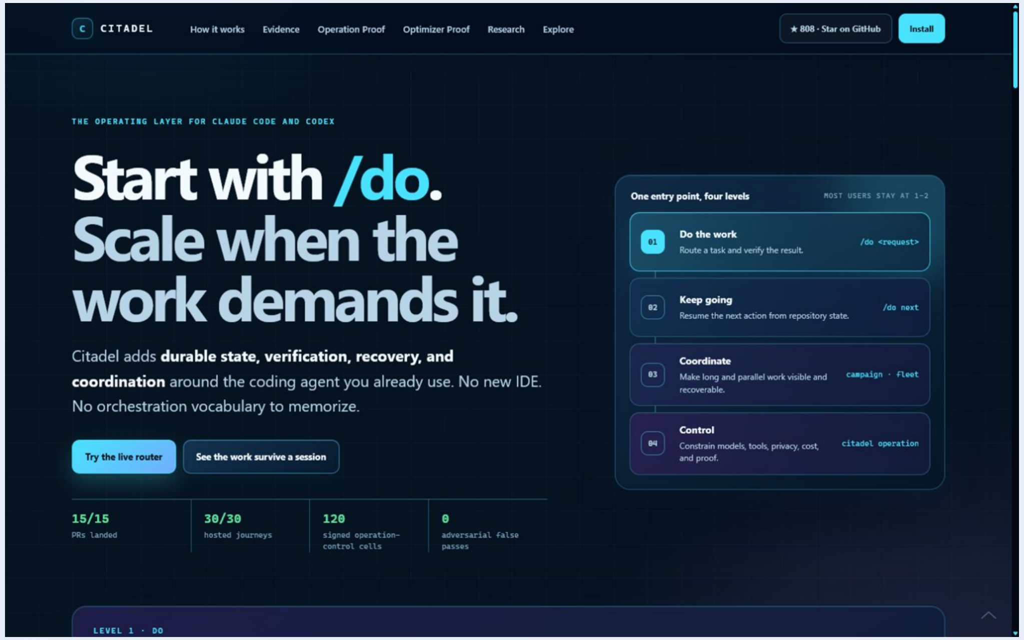
Models, tools, topology, retries, privacy, time, cost, and stop rules.

RUN — **Runtime coverage**

Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.

PROOF — **Evidence plane**

External artifact verdicts, failure preservation, signed receipts, and offline replay.



PINNED SENTIENT ROMA DIAGNOSTIC

ROMA can emit reconstructable Citadel evidence.

The frozen 24-cell diagnostic retired integration and proof risk while exposing the controller work the grant must fund.

24 / 24
cells measured

0
false passes

0
control or integrity failures

4
policies compared

RISK RETIRED

- Pinned ROMA execution emitted requested and observed route identity.
- A model-external artifact verifier decided completion.
- Signed receipts reconstruct offline without a Citadel service.

FUNDED RISK

The initial Citadel + ROMA heuristic verified 4/6 cells in 1042.8s versus frontier 6/6 in 47.7s. It avoided no strong whole-operation attempts.

NO SAVINGS CLAIM - GATE FAILED

Evidence: benchmarks/roma-operation-control/published-run/REPORT.md | Verify: npm run operation-proof:verify

BUILT BEFORE FUNDING

One maintainer already shipped the substrate.

Citadel is a public system with continuing repository pull, not a grant-funded mockup.

134
days public

496 / 520
main commits attributed to Seth

808 / 79
stars / forks

524
unique cloners in 14 days

WORKING SUBSTRATE

- Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- 49 workflows and 35 hooks across 29 lifecycle events.
- Windows, Linux, and macOS verification on every release path.

CLAIM DISCIPLINE

Three frozen savings stories were rejected under the same public evidence contract. Those failures define the controller requirements instead of being marketed away.

FUNDING SCALES A WORKING BASE

GitHub observed 2026-08-01. Clone window: 2026-07-18 to 2026-07-31. Clones are not installs, users, or adoption.

NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

Funding buys the unproven result.

Scale a working control and proof layer into a representative optimizer - or publish a reusable negative result.

01	**Representative workload**	60+ artifact-producing operations and complete cost boundary	$30k
02	**Adapter SDK**	Sentient ROMA plus two additional open stacks	$28k
03	**Expected-value controller**	Price verification, escalation, recovery, and failure	$35k
04	**Prospective evaluation**	Frozen multi-stack policy and statistical contract	$32k
05	**Public proof and release**	Hosted reconstruction, /do onboarding, support window, and maintenance	$25k

Canonical milestones and acceptance gates: docs/grants/MILESTONES_AND_BUDGET.md

NO-GO RULES ARE PART OF THE PRODUCT

Capped spend. Hard outcome gates.

COST BASIS

Maintainer labor	**$99,000**
Evaluation compute and tools	**$27,000**
Hosted verification and retention	**$7,000**
Reference hardware and measurement	**$7,000**
Compatibility and accessibility costs	**$2,000**
Bounded contingency	**$8,000**
Total	**$150,000**

PRIMARY PROSPECTIVE GATE

>= 80%
absolute verified completion

>= 95%
of valid frontier rate

>= 30%
lower measured end-to-end cost

ZERO
false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

Tranches: $45k / $60k / $45k. Compute credits reduce cash draw dollar-for-dollar; unused funds remain unspent.

DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

An open optimizer that can prove when it is wrong.

WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- Signed evidence lowers substrate risk and exposes the remaining research risk.

WHAT STAYS PUBLIC

- MIT code, adapters, contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

Seth Gammon - solo builder and maintainer

OPEN. VERIFIABLE. FALSIFIABLE.

Evaluator: github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md